

Designing and fitting models: A Zombie Apocalypse

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Recap: Euler-Binomial Scheme

We previously considered continuous time models with a simple, one-option flow:

$$S \rightarrow E \rightarrow I \rightarrow R$$

We did this by turning continuous transition rates μ_{SE} into transition probabilities over a small time step dt :

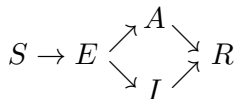
$$N_{SE}(t) \sim \text{Binom}(S(t), 1 - e^{-\mu_{SE} \times dt})$$

```
double p = 1 - exp(-mu_SE * dt);  
double trans_SE = rbinom(S, p); // Draw from Susceptible pool  
S -= trans_SE;  
E += trans_SE;
```

This works perfectly when there is only **one** way out of a compartment.

The Problem: Competing Risks

What happens when an individual can exit a compartment in multiple ways? Consider a model where an exposed individual may become either **symptomatic** (I), or **asymptomatic** (A):



If we use sequential `rbinom` draws, **the first draw always gets priority**, creating a mathematical bias:

```
double trans_I = rbinom(E, p_I);  
// Bias! Asymptomatic draw is artificially restricted by the first draw  
double trans_A = rbinom(E - trans_I, p_A);
```

Euler-Multinomial Distribution

To avoid this bias, we need to draw both transitions at the same time. We can handle convert rates into probabilities, and do simultaneous draws, with a single distribution:
Euler-Multinomial.

This is a natural extension of the process of converting continuous-time rates to probabilities, and doing binomial draws, which has two options: staying in E , or going to I .

Now, we may have 3 options: Staying in E , going to I , or going to A :

$$(N_{EI}(t), N_{EA}(t)) \sim \text{EulerMultinom}[E(t), (\mu_{EI}(t), \mu_{EA}(t))]$$

reulermultinom in practice

The `reulermultinom` function does this for us. It calculates the total exits based on the sum of all rates, then correctly distributes them.

Correct: `reulermultinom`

```
// 1. Arrays for rates and output
double rates[2] = {mu_EI, mu_EA};
double trans[2];

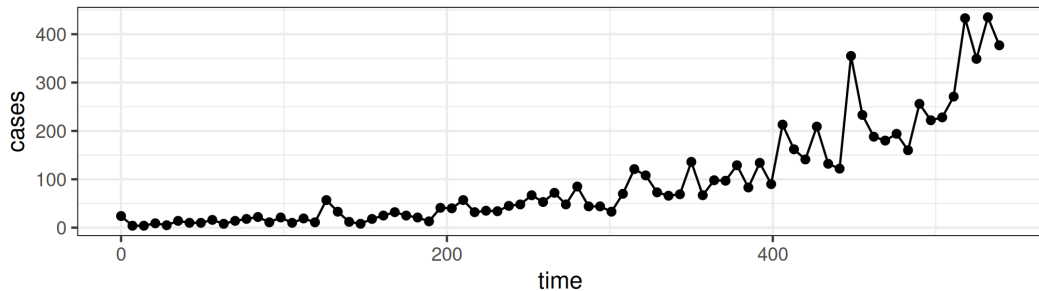
// 2. Draw simultaneously
// (num_paths, size, &rates, dt, &trans)
reulermultinom(2, E, &rates[0], dt, &trans[0]);

// 3. Extract results
double trans_I = trans[0];
double trans_A = trans[1];
```

Note: The `&` symbol just tells C exactly where your lists are stored in memory.

Introduction I

- ▶ Our goal: practice what we learned, design a model and make predictions.
- ▶ **Scenario:** A zombie-apocalyptic infection has spread on a small island nation.
- ▶ Simulated data available on course website: weekly data for 1.5 years



Goals

- ▶ Fit a model
- ▶ Estimate number of surviving humans, forecasts, etc.

The Anomaly

- ▶ **The Setup:** Public health officials are tracking a bizarre neurological outbreak (colloquially “Zombies”).
- ▶ **The Hypothesis:** The pathogen circulated completely undetected for 2 months ($t < 60$) causing no reportable symptoms. At Day 60, it mutated.
- ▶ **The Mutation:** A high (unknown) percentage of exposed individuals turn into aggressive Zombies; the remaining proportion develop a rash and become permanently immune.
- ▶ **Spread:** The virus spreads from contacts between current zombies + exposed individuals and susceptible survivors.
- ▶ **Containment:** Zombies do not die naturally. They are destroyed via mass-action combat driven by surviving humans (Susceptible + Immune).

The Fog of War (Measurement Model)

- ▶ We only partially observe total zombies, though reporting is high.
- ▶ The exposed class (those that may become zombies) is completely invisible.
- ▶ Reporting infrastructure decays (exponentially) as the total zombie horde (Z) grows.

Model building

- ▶ What compartments should we include based on what we know?
- ▶ New infections follow similar dynamics relative to what we have seen. What about the mass-action combat destruction of zombies?
- ▶ Modeling and forecasting long periods. Should we include births / deaths?

Independent Exercise: Map the Competing Risks (15 mins)

On Paper or Whiteboard: Map the transition flow for this system. Pay special attention to competing risks (arrows leaving the same box).

1. What are the two exit pathways for a Susceptible (S) individual?
2. What are the three exit pathways for an Exposed (E) individual?
3. If R holds dead bodies and destroyed zombies, do they contribute to the birth rate (μ_b)?
4. Should R be in the transmission denominator (N)? Why or why not?

Additional C-Snippet Hints

- ▶ **The Mutation Time:** The mutation doesn't unlock until 60 days post-infection. You can use a time condition in C: `double rate_E_out = (t < t_mut) ? 0.0 : gamma;`
- ▶ **Data Offset:** Our data starts exactly when the zombies appear (Week 0). However, Patient Zero was infected roughly 60 days *before* this. Consider how to set `t0` in your `pomp` object to allow the unobserved E compartment to grow before the data begins.
- ▶ **Incidence Tracking:** We only observe *new* cases. Make sure to accumulate these (`Z_inc`) properly.

Independent Exercise: Build the Model (45 mins)

Your Task: Work in your groups to translate your conceptual model into pomp C-snippets.

1. Write `rinit`, `rprocess`, `rmeasure`, and `dmeasure` snippets.
2. Build the pomp object.
3. Choose a reasonable set of starting parameters and use `simulate()` to ensure your model produces an outbreak curve that roughly matches the shape of the data.

Estimation with Iterated Filtering

The dataset contains a massive gap: the first two months of the outbreak are entirely unobserved.

The Challenge: When you run `mif2`, the particles must correctly estimate the early, silent transmission rate (β_E) based *only* on the magnitude of the zombie explosion that happens weeks later.

Independent Exercise: Estimation & The Doomsday Forecast (60 mins)

1. **Estimation:** Run `mif2` to estimate the transmission rates (β_E, β_Z), the combat effectiveness (α), and the immunity fraction (ω). Plot your convergence traces.
2. **Forecasting:** Extract your best parameter estimates. Use `simulate()` to generate 50 forward trajectories extending 10 years into the future.
3. **Discussion:** Look at your simulated trajectories for the Susceptible (S) compartment. Does the human-driven combat mechanic allow humanity to bounce back, or does the dropping population trigger a runaway collapse?